MIL-C-51074C
29 June 1967
SUPERSEDING
MIL-C-51074B(MU)
2 November 1966

MILITARY SPECIFICATION

CALCULATOR, RADIAC, ABC-M1

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of radiological calculator. The calculator is a circular type slide rule consisting of three concentric disks containing the applicable scales.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

DL124-13-3	-	Calculator, Radiac, ABC-M1.

(All drawings referenced in this specification are listed on Edgewood Arsenal Drawing DL124-13-3.)

FSC 6665

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:49Z
Check the source to verify that this is the current version before use.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL124-13-3 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings. The ink used for marking and lettering shall be compatible with the vinyl chloride material. The printing shall be legible, nonfading, and durable.

3.1.2 Components. All components of the calculator shall conform to the specifications and drawings listed on Drawing DL124-13-3 and subsidiary drawings.

3.2 Manufacture and assembly. The calculator shall be manufactured and assembled in accordance with Drawing C124-13-3.

3.3 Functioning. The calculator, when tested in accordance with 4.3.4, shall be capable of solving the problems for total dose using the data listed in table I.

3.4 Preproduction. Prior to the start of regular production, a preproduction sample of ABC-M1 calculators shall be produced in accordance with this specification for examination and test (see 4.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:49Z
Check the source to verify that this is the current version before use.

Table I. Problem data

Scale	Data		
	Problem 1	Problem 2	Problem 3
Time after burst	1 hour	10 hours	7 hours
Dose rate	30 rads/hour	500 rads/hour	6 rads/hour
Time of entry	40 minutes	2 days	10 hours
Time of exit	10 hours	2 weeks	20 hours
Total dose	69 ± 2 rads	5900 ± 200 rads	25 ± 2 rads

3.5 Workmanship. The calculator shall be free from imperfections (pits, bumps, cracks, flash, abrasions, blisters, holes, or porous areas) and contamination (dirt, grease, oil or foreign matter). All surfaces shall be smooth, and laminations shall be uniform without unsealed areas.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:49Z
Check the source to verify that this is the current version before use.

4.2 Preproduction.

4.2.1 Sample. A preproduction sample of five completely assembled ABC-M1 calculators and five unassembled ABC-M1 calculators shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.2.2 Examination and nondestructive test. The components of the five unassembled calculators shall be examined for all the requirements of the drawings. The five assembled calculators shall be examined and tested in accordance with the classification of defects (4.3.3.2).

4.2.3 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.2.2 to be acceptable.

4.3 Regular production.

4.3.1 Lotting. A lot shall consist of the ABC-M1 calculators produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

4.3.2 Sampling for examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105. For Major defect 101, level S-2 shall be used.

4.3.3 Inspection.

4.3.3.1 Examination and nondestructive tests. Sample calculators shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.3.3.2 Classification of defects.

(a) Calculator, radiac, ABC-M1 (Dwg C124-13-3).

Categories	Defects	Acceptance standards
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Functioning	4.3.4

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:49Z
Check the source to verify that this is the currentversion before use.

4.3.3.2 Classification of defects (continued).

(a) Calculator, radiac, ABC-M1 (Dwg C124-13-3).

Categories	Defects	Acceptance standards
Major:	AQL 2.5 percent defective	
102	Component missing, incorrect, or incorrectly assembled	
103	Disks not free to turn easily	
104	Marking or lettering missing, incorrect or illegible; or color incorrect	
105	Rivet holes in disks incorrectly located	
106	Workmanship (see 3.5)	

4.3.4 Functioning test. Test shall be conducted as follows:

(a) Problem 1. Align a reading of one hour on the intermediate disk opposite a reading of 30 rads/hour on the outer disk. Align a reading of 40 minutes on inner disk opposite the "time of entry" arrow on the intermediate disk. From a reading of 10 hours on inner disk, follow the colored guide band to the outer disk and read the total dose on the outer disk opposite the same relative relationship on the colored guide band.

(b) Problem 2. Align a reading of 10 hours on the intermediate disk opposite a reading of 500 rads/hour on the outer disk. Place a reading of 2 days on inner disk opposite the "time of entry" arrow on the intermediate disk. From a reading of 2 weeks on the inner disk, follow the colored guide band to the outer disk and read the total dose on the outer disk opposite the same relative relationship on the colored guide band.

(c) Problem 3. Align a reading of 7 hours on the intermediate disk with a reading of 6 rads/hour on the outer disk. Align a reading of 10 hours on inner disk opposite the "time of entry" arrow on the intermediate disk. From a reading of 20 hours on the inner disk, follow the colored guide band to the outer disk and read the total dose on the outer disk opposite the same relative relationship on the colored guide band.

Source: https://assist.dla.mil -- Downloaded: 20261O-22T20:49Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level C. The calculators shall be packaged to provide adequate protection against deterioration and damage from the supply source to the first receiving activity for immediate use.

5.2 Packing, level C. Calculators, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The M1 radiac calculator is intended for use in radiological warfare for calculating past, present or future dose rates, total dose, and entry and exit times.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Preproduction:

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time allowed for the Government to notify the supplier whether or not to proceed with production.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:49Z
Check the source to verify that this is the current version before use.

6.3 International interest. Certain provisions of this specification are the subject of international standardization agreement (see Serial No. 27, Supplement B to Quadripartite Standardization List Sheet Report). When amendment, revision or cancellation of this specification is proposed which will affect or violate the international agreement concerned, the preparing activity will take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

Custodians:

Army - MU
Navy - SH
Air Force - 82

Preparing activity:

Army - MU (EA)
Project No. 6665-0153

International interest (see section 6)

Coordination Method Code: C

Review activities:

Air Force - 82
Navy - SH, AS, OS

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:49Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:49Z
Check the source to verify that this is the current version before use.